Shardora: Structural Privacy-Preserving Transactions in Sharded Blockchains via Observation Domain Separation and Autonomous Execution

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***Abstract*—Privacy-preserving transactions on monolithic blockchain state machines face three insurmountable structural bottlenecks: (1) Gas linkage attacks, where the imperative of gas payment unavoidably compromises the funding trails of unlinked recipient addresses; (2) Global observation domain collapse, where universal validator visibility enables pervasive timing, metadata, and graph-theoretic correlation; and (3) Trust expansion, where cross-domain operations require reliance on external relayers, centralized sequencers, or multi-party computation (MPC) quorums.**

In this paper, we formalize the structural privacy advantages of sharded architectures and present the complete protocol specification of Shardora. We prove an impossibility result (Theorem A) demonstrating that under standard Gas-First Account State Machines (GASM), simultaneous sender privacy, recipient gas autonomy, and decentralization cannot be attained without protocol-level autonomous execution (APLE). We then formalize two foundational cross-shard privacy properties: Physical Observation Domain Separation (Property N1) and Zero-Trust-Increment Cross-Shard Authorization (Property N2). We prove that under ideal cross-shard synchronization, the target shard observer's linking advantage is information-theoretically bounded by $1/K$ (Theorem B), while cryptographic confidentiality and ledger integrity decouple into mutually independent reductions (Theorem D). Furthermore, we model real-world network side channels (packet timing, Global Passive Adversaries, and amount graph analyses) and establish the real-world degradation bound (Theorem F), demonstrating that a lightweight mitigation stack bounds adversary advantage to $\leq 1/K + 10^{-3}$. Target shards eliminate bilinear pairings entirely via Cross-Shard Credit Certificates (CSCC), yielding zero-pairing blind commitment minting and achieving an estimated analytical throughput scaling up to 16,000–32,000 TPS across shards.

***Index Terms*—Blockchain Sharding, Zero-Knowledge Proofs, Transaction Privacy, Cross-Shard Consensus, Impossibility Results, Side-Channel Analysis.**

I. Introduction

A. Core Thesis: Structural vs. Performance Dividends of Sharding

Privacy-preserving transactional systems on transparent ledgers have predominantly evolved within single-chain paradigms. Implementations such as Zerocash [1], Tornado Cash [2], and Aztec [6] demonstrate that zero-knowledge succinct non-interactive arguments of knowledge (zk-SNARKs) can successfully shield transaction amounts and witness paths. However, deploying privacy systems upon a globally visible, monolithic state machine exposes fundamental structural limitations that cannot be resolved solely through stronger cryptographic primitives.

Consensus in single-chain architectures enforces a total ordering across all nodes. Consequently, every validator maintains a globally comprehensive view of state updates. While zero-knowledge cryptography successfully hides witness data, it cannot obscure the existence, block height, and temporal arrival of transactions. In contrast, a sharded blockchain natively partitions both the state space and the validator quorum. This paper demonstrates that a multi-shard architecture yields **structural privacy dividends** that solve challenges inherent to single-chain systems, rather than merely providing horizontal performance scaling.

Specifically, we prove that cross-shard architectures overcome three structural glass ceilings that single-chain systems cannot evade:

1) **Elimination of Gas Linkage:** By delegating target execution to an internal protocol-level execution entity, recipient accounts with zero balance receive shielded funds without leaking ancestral funding paths.
2) **Architectural Observation Domain Separation:** Disjoint validator quorums physically partition the adversary's visibility, turning transaction linking into an information-theoretically bounded problem.
3) **Zero-Trust-Increment Authorization:** Cross-shard consensus certificates allow destination shards to authorize balance updates based strictly on underlying consensus safety, avoiding the need for external bridging trust assumptions.

B. Problem I: Gas Linkage Attacks

In an account-based state machine, any transaction execution requires an upfront gas fee deduction from the initiator's balance. In privacy mixers (e.g., Tornado Cash), a withdrawal must be submitted from a newly generated address possessing zero native token balance. To fund the gas fee, users face an unresolvable trilemma:

- **Direct Funding:** Funding the new address from an existing account immediately creates a deterministic on-chain link, destroying sender anonymity.
- **Third-Party Relayers:** Offloading gas payment to centralized relayers introduces IP/timing metadata leakage and fee-skimming vulnerabilities.
- **Private Mempools:** Mechanisms like Flashbots or account abstraction (ERC-4337) merely displace the correlation

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point onto specialized bundlers or paymasters, which remain globally identifiable aggregators.

This limitation is structural: single-chain state machines mandate that transaction dispatchers possess balance prior to execution.

C. Problem II: Observation Domain Collapse

Single-chain consensus mandates that every validator processes every transaction. Although zero-knowledge proofs hide internal witness data, they cannot obscure the external existence, block timestamp, and boundary conditions of shielded deposits and withdrawals. A passive global observer observes a unified "black box" entrance and exit timetable, where temporal correlations and volume spikes can be exploited via statistical traffic analysis. This cannot be resolved by cryptographic compression, as consensus requires full broadcast.

D. Problem III: Trust Expansion in Cross-Chain Authorization

Extending privacy cross-chain typically requires collateral lock-and-mint semantics. Authorization for state transition on the target chain must be granted by an intermediary entity: centralized sequencers (e.g., Aztec v2), independent multi-party computation (MPC) quorums, or external relayer networks. These mechanisms introduce disjoint security assumptions, compounding attack surfaces beyond the underlying layer-1 consensus.

E. Contributions

In this work, we demonstrate that a multi-shard blockchain architecture provides structural privacy advantages that natively circumvent the structural impossibility results of single-chain privacy. Our core theoretical and architectural contributions are:

1) **Impossibility Result (Theorem A):** We formalize the Gas-First Account State Machine (GASM) and prove that no protocol on a GASM can simultaneously achieve sender privacy, recipient gas autonomy, and decentralization. We prove that Shardora evades this impossibility via Autonomous Protocol-Level Execution (APLE).
2) **Formalization of Properties N1 and N2:** We define Physical Observation Domain Separation (N1) and Zero-Trust-Increment Authorization (N2). We construct a cross-shard transaction mechanism satisfying both properties simultaneously.
3) **Information-Theoretic Source Unlinkability (Theorem B):** We prove that in an idealized cross-shard model where shard validator sets are mutually disjoint ($\mathcal{V}_{\text{src}} \cap \mathcal{V}_{\text{dst}} = \emptyset$), the target shard's linking advantage regarding the source nullifier is information-theoretically bounded by $1/K$, independent of adversary computational bounds.
4) **Dual-Track Decoupled Security (Theorem D):** We demonstrate that Note confidentiality reduces strictly to the computational Diffie-Hellman (CDH) hardness of the recipient's private key, while ledger integrity reduces to BFT consensus safety and BLS existential unforgeability. Hence, a total collapse of BFT consensus leaves historical Note secrecy intact.
5) **Real-World Network Degradation Bound (Theorem F):** We model timing side channels, Global Passive Adversaries (GPA), and amount graph analysis. We formulate an explicit degradation bound and prove that a lightweight mitigation stack bounds adversary advantage to $\leq 1/K + 10^{-3}$.
6) **Zero-Pairing Blind Target Minting:** Target shards process incoming shielded commitments with zero pairing operations ($\mathcal{O}(1)$ BLS verification) via Cross-Shard Credit Certificates (CSCC), eliminating the verification pairing bottleneck.

F. Compliance-Friendly Positioning

Following the regulatory landscape established after the OFAC sanctions on Tornado Cash in 2022, privacy protocols must address compliance risks where malicious actors share anonymity sets with legitimate users. Shardora supports optional regulatory compliance via the Privacy Pools paradigm [3]. Within the Zero-Knowledge circuit, an optional constraint C11 proves non-membership in an on-chain sanctions accumulator ($\texttt{old_note} \notin \text{MerkleTree}(\mathcal{B}_{\text{sanction}})$). This constraint is completely orthogonal to Properties N1 and N2; removing C11 leaves the structural properties intact, while enabling it allows users to establish association-set compliance without compromising transactional confidentiality.

II. Related Work and Comparative Landscape

A. Taxonomy of Existing Solutions

We systematically categorize existing privacy solutions into six technological paradigms, evaluating their operational profiles against cross-shard requirements:

- **Monolithic zk-SNARK Mixers (Tornado Cash [2], Privacy Pools [3]):** Enforce equal-denomination pools on Ethereum. Deposits and withdrawals occur on the same chain; all full nodes construct a global transaction graph. Gas linkage attacks remain structural.
- **Privacy-Centric L1s (Zcash Sapling/Orchard [4], Monero [5]):** Native shielded pools eliminate address leakage within the native chain, but lack EVM compatibility and cannot route value across heterogeneous shards or independent chains without plaintext bridges.
- **ZK-Rollup L2s (Aztec v2 [6]):** Shield transactions off-chain, submitting aggregate state roots to Ethereum L1. However, L1 deposit and withdraw events remain public, and the centralized sequencer acts as a singular correlation and censorship point.
- **IBC-Enabled Shielded Pools (Penumbra, Namada):** Enable cross-chain privacy via Inter-Blockchain Communication (IBC). However, IBC packet headers (source, destination, sequence, amount) traverse relayers in plaintext, violating N1 and N2.
- **Lock-and-Mint ZK Bridges (Railgun, zkBridge):** Use public lock events on the source chain paired with mint events on the target chain. The lock/mint pairs expose temporal and value correspondences on public ledgers.

- **Confidential Smart Contract Platforms (Secret Network, Oasis):** Rely on Trusted Execution Environments (TEEs), introducing hardware side-channel risks and centralized enclave vendor dependencies.

B. Formal Differentiation of N1 and N2

Definition 1 (Property N1: Physical Observation Domain Separation)**.** Let $\mathcal{V}_{\text{src}}$ and $\mathcal{V}_{\text{dst}}$ denote the validator quorums of the source and target shards, respectively. A protocol satisfies Property N1 if $\mathcal{V}_{\text{src}} \cap \mathcal{V}_{\text{dst}} = \emptyset$, and for any probabilistic polynomial-time (PPT) adversary $\mathcal{A}$ corrupting sub-threshold quorums $|\mathcal{C}_{\mathcal{A}} \cap \mathcal{V}_{\text{src}}| < \lceil 2n/3 \rceil$ and $|\mathcal{C}_{\mathcal{A}} \cap \mathcal{V}_{\text{dst}}| < \lceil 2n/3 \rceil$, the ledger view of the target shard View_{dst} is statistically independent of the spent source nullifiers:

$$I(\text{nullifier}_{\text{src}}; \text{View}_{\text{dst}}) = 0. \tag{1}$$

Definition 2 (Property N2: Zero-Trust-Increment Authorization)**.** Let $\mathcal{T}_{\text{base}}$ be the set of trust assumptions required for base layer BFT consensus (i.e., $< 1/3$ Byzantine nodes per shard). A cross-shard privacy protocol satisfies Property N2 if its cross-shard mint authorization trust assumption $\mathcal{T}_{\text{auth}}$ satisfies:

$$\mathcal{T}_{\text{auth}} \subseteq \mathcal{T}_{\text{base}} \implies \mathcal{T}_{\text{total}} = \mathcal{T}_{\text{base}}. \tag{2}$$

Table I presents a systematic evaluation of contemporary privacy architectures against these criteria.

C. Comparative Analysis: Multi-Pool Bridges vs. Shardora

A common counter-proposal to native sharded privacy is chaining multiple independent mixing pools across heterogeneous chains using standard bridging infrastructure. We highlight two primary issues with this approach:

1) **Breakdown of Property N1:** In a bridged configuration, a lock event E_{lock} on Chain A corresponds deterministically to a mint event E_{mint} on Chain B. For unique transfer amounts, $\Pr[\text{link}(E_{\text{lock}}, E_{\text{mint}})] = 1$. Even with standardized denominations, the bridging transaction leaks timing and relayer metadata, reducing the anonymity set to the concurrent bridging volume $N_\delta \ll K$.
2) **Violation of Property N2:** Bridges rely on independent multisig signers or MPC networks. Compromises in bridge committees (e.g., Ronin, Wormhole, Nomad) directly compromise cross-chain asset backing without breaching base-layer consensus.

III. System Architecture and Protocol Design

A. System Abstractions and Protocol Entities

To evaluate cross-shard privacy formally, we map Shardora's implementation constructs to standardized academic protocol primitives:

- **Cross-Shard Shielded Carrier (CSSC):** The cross-shard message carrier (`ToTxMessageItem`) encapsulating a blind commitment, an ECIES ciphertext, and a BLS consensus signature. It contains no plaintext addresses or amounts.
- **Zero-Pairing Blind Commitment Minting (ZPCM):** The execution logic (`ShieldedCreditFromCSCC`) on the target shard that validates the source BFT certificate and inserts the blind commitment into the target Merkle tree without running bilinear pairing operations or decrypting payloads.
- **Autonomous Protocol-Level Execution (APLE):** The internal state-machine executor (`SYSTEM_EXECUTOR`, address `0x...ff`) instantiated by the target consensus engine. APLE executes state transitions with zero gas cost charged to the recipient.
- **Decentralized Shielded Pool Engine (DSPE):** The smart contract state machine (`PrivacyShadow`) managing the Sparse Merkle Tree (SMT) of commitments, spent nullifier sets, and denomination pools within each shard.

B. Cryptographic Primitives and Infrastructure Reuse

Shardora exploits the existing elliptic curve infrastructure over the BN254 (`alt_bn128`) curve:

- **Curve Groups:** $\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T$ with pairing $e : \mathbb{G}_1 \times \mathbb{G}_2 \to \mathbb{G}_T$ and prime order $p \approx 2^{254}$. Independent generators $G, H \in \mathbb{G}_1$ such that $\log_G(H)$ is unknown.
- **Pedersen Commitments:** A Note with value $v \in [0, 2^{64}-1]$ and blinding scalar $r \xleftarrow{\$} \mathbb{F}_p$ is committed as $\text{cm} = v \cdot H + r \cdot G \in \mathbb{G}_1$.
- **Poseidon Hash Function:** A SNARK-friendly algebraic hash function $H_{\text{p}} : \mathbb{F}_p^* \to \mathbb{F}_p$ instantiated over the BN254 scalar field.
- **Groth16 zk-SNARK:** Proving relations over $\mathbb{F}_p$ with proof $\pi \in \mathbb{G}_1^2 \times \mathbb{G}_2$, verified via 3 pairing evaluations on the source shard.
- **Threshold BLS Signatures:** Reused from base HotStuff/zBFT consensus to generate cross-shard credit certificates without new keys.

C. Pure Commitment Ledger and Stealth Addresses

A user maintains a spending key $\texttt{sk} \in \mathbb{F}_p$ and a viewing private key $\texttt{vsk} \in \mathbb{F}_p$. The public counterparts are published as:

$$\texttt{pk}_{\text{spend}} = \texttt{sk} \cdot G, \quad \texttt{pk}_{\text{view}} = \texttt{vsk} \cdot G \quad \in \mathbb{G}_1. \tag{3}$$

For each outbound transfer, the sender samples an ephemeral secret $r_{\text{eph}} \xleftarrow{\$} \mathbb{F}_p$, computes ephemeral public key $R = r_{\text{eph}} \cdot G$, and calculates the ECDH shared point $S = r_{\text{eph}} \cdot \texttt{pk}_{\text{view}}$. The one-time spending public key is derived as:

$$\texttt{pk}_{\text{stealth}} = H_{\text{p}}(S) \cdot G + \texttt{pk}_{\text{spend}}. \tag{4}$$

$\texttt{pk}_{\text{stealth}}$ is never published in plaintext on-chain. Shardora maintains a **Pure Commitment Ledger**: the on-chain ledger records only the commitment cm and ciphertext $\texttt{ct}_{\text{ecies}}$. The stealth address remains internal to the Note plaintext. To prevent double-spending without revealing the underlying commitment, the nullifier is defined as:

$$\texttt{nul} = H_{\text{p}}(\texttt{sk}_{\text{note}} \parallel \text{cm}), \tag{5}$$

where $\texttt{sk}_{\text{note}} = H_{\text{p}}(S) + \texttt{sk}$.

D. Zero-Knowledge Circuit Constraints

The Groth16 circuit enforces validity of Note consumption and generation:

- **Public Inputs:** $\texttt{nul}$, $\texttt{root}_{\text{old}}$, cm_{send}, $\text{cm}_{\text{change}}$, and value binding $\texttt{bind} = v_{\text{send}} \cdot H$.
- **Private Witnesses:** $\texttt{note}_{\text{old}} = (v_{\text{old}}, r_{\text{old}}, \texttt{pk}_{\text{old}})$, Merkle path $\mathcal{P}$, $\texttt{sk}_{\text{old}}$, $v_{\text{send}}, r_{\text{send}}, \texttt{pk}_{\text{stealth}}$, and $v_{\text{change}}, r_{\text{change}}$.

The Rank-1 Constraint System (R1CS) enforces relations C1–C10:

$$\textbf{C1:} \quad \text{cm}_{\text{old}} = v_{\text{old}} \cdot H + r_{\text{old}} \cdot G \tag{6}$$

$$\textbf{C2:} \quad \text{MerkleVerify}(\texttt{root}_{\text{old}}, \text{cm}_{\text{old}}, \mathcal{P}) = 1 \tag{7}$$

$$\textbf{C3:} \quad \texttt{nul} = H_{\text{p}}(\texttt{sk}_{\text{old}} \parallel \text{cm}_{\text{old}}) \tag{8}$$

$$\textbf{C4:} \quad \texttt{sk}_{\text{old}} \cdot G = \texttt{pk}_{\text{old}} \tag{9}$$

$$\textbf{C5:} \quad v_{\text{old}} = v_{\text{send}} + v_{\text{change}} \tag{10}$$

$$\textbf{C6, C7:} \quad v_{\text{send}} \in [0, 2^{64} - 1], \quad v_{\text{change}} \in [0, 2^{64} - 1] \tag{11}$$

$$\textbf{C8:} \quad \text{cm}_{\text{send}} = v_{\text{send}} \cdot H + r_{\text{send}} \cdot G \tag{12}$$

$$\textbf{C9:} \quad \text{cm}_{\text{change}} = v_{\text{change}} \cdot H + r_{\text{change}} \cdot G \tag{13}$$

$$\textbf{C10:} \quad \texttt{bind} = v_{\text{send}} \cdot H \tag{14}$$

Table III summarizes the sub-circuit constraint complexity, totaling 21,010 constraints.

E. Cross-Shard Payload: ECIES and CSCC

The sender encrypts the plaintext tuple $\texttt{pt} = (v_{\text{send}}, r_{\text{send}}, \texttt{pk}_{\text{stealth}})$ under the recipient's viewing public key $\texttt{pk}_{\text{view}}$:

$$R = r_{\text{eph}} \cdot G \in \mathbb{G}_1, \quad S = r_{\text{eph}} \cdot \texttt{pk}_{\text{view}}, \tag{15}$$

$$k_{\text{enc}} = \text{HKDF-SHA256}(S \parallel \text{"}\texttt{shardora_note_v1}\text{"}), \tag{16}$$

$$\texttt{ct} = \text{AES-256-GCM-Enc}_{k_{\text{enc}}}(\texttt{pt}, \text{AAD} = \text{cm}_{\text{send}}), \tag{17}$$

$$\texttt{ct}_{\text{ecies}} = R \parallel \texttt{ct} \quad (\approx 144 \text{ Bytes}). \tag{18}$$

Upon verifying Groth16 proof π inside the source shard EVM, the source BFT committee attaches a BLS threshold signature over the cross-shard transfer request during block finalization:

$$\sigma_{\text{cscc}} = \text{BLS-Sign}(\texttt{sk}_{\text{src_bft}}, H(\text{cm}_{\text{send}} \parallel \texttt{shard}_{\text{dst}} \parallel \texttt{pool}_{\text{dst}} \parallel \texttt{hash}_{\text{blk}})) \tag{19}$$

The cross-shard carrier message CSSC contains:

$$\text{CSSC} = \langle \text{cm}_{\text{send}}, \texttt{ct}_{\text{ecies}}, \sigma_{\text{cscc}}, \texttt{hash}_{\text{blk}}, \texttt{shard}_{\text{dst}}, \texttt{pool}_{\text{dst}} \rangle. \tag{20}$$

The message size is bounded to ≈ 354 Bytes, containing no plaintext values or user addresses.

F. Execution Flow and Protocol Algorithms

The complete end-to-end execution of cross-shard shielded transfers spans client dispatch, source shard attestation, message routing, and target execution, formalized in Algorithms 1 and 2.

Algorithm 1 Client Generation and Source Shard Submission

Require: Sender keys $(\texttt{sk}, \texttt{vsk})$, Note $\texttt{note}_{\text{old}}$, Recipient keys $(\texttt{pk}_{\text{spend}}, \texttt{pk}_{\text{view}})$, target shard $\texttt{shard}_{\text{dst}}$, pool $\texttt{pool}_{\text{dst}}$, transfer amount v_{send}.
Ensure: Shielded transaction submission $\texttt{tx}_{\text{shielded}}$.

1: Sample ephemeral secret $r_{\text{eph}} \xleftarrow{\$} \mathbb{F}_p$.
2: Compute $R \leftarrow r_{\text{eph}} \cdot G$, $S \leftarrow r_{\text{eph}} \cdot \texttt{pk}_{\text{view}}$.
3: Derive $\texttt{pk}_{\text{stealth}} \leftarrow H_{\text{p}}(S) \cdot G + \texttt{pk}_{\text{spend}}$.
4: Sample blinding scalars $r_{\text{send}}, r_{\text{change}} \xleftarrow{\$} \mathbb{F}_p$.
5: Compute $v_{\text{change}} \leftarrow v_{\text{old}} - v_{\text{send}}$.
6: Construct $\texttt{note}_{\text{send}} \leftarrow (v_{\text{send}}, r_{\text{send}}, \texttt{pk}_{\text{stealth}})$.
7: Construct $\texttt{note}_{\text{change}} \leftarrow (v_{\text{change}}, r_{\text{change}}, \texttt{pk}_{\text{spend}})$.
8: Generate Pedersen commitments $\text{cm}_{\text{send}}, \text{cm}_{\text{change}}$.
9: Compute nullifier $\texttt{nul} \leftarrow H_{\text{p}}(\texttt{sk}_{\text{old}} \parallel \text{cm}_{\text{old}})$.
10: Generate Groth16 zk-SNARK proof π for R1CS relations C1–C10.
11: Compute $k_{\text{enc}} \leftarrow \text{HKDF-SHA256}(S \parallel \text{"}\texttt{shardora_note_v1}\text{"})$.
12: Compute $\texttt{ct} \leftarrow \text{AES-256-GCM-Enc}_{k_{\text{enc}}}(\texttt{note}_{\text{send}}, \text{cm}_{\text{send}})$.
13: Set $\texttt{ct}_{\text{ecies}} \leftarrow R \parallel \texttt{ct}$.
14: Dispatch $\texttt{tx}_{\text{shielded}} \leftarrow (\texttt{nul}, \text{cm}_{\text{send}}, \text{cm}_{\text{change}}, \pi, \texttt{ct}_{\text{ecies}}, \texttt{shard}_{\text{dst}}, \texttt{pool}_{\text{dst}})$ to source shard DSPE.

Algorithm 2 Target Shard ZPCM Autonomous Execution (APLE)

Require: Arriving carrier message $\text{CSSC} = \langle \text{cm}_{\text{send}}, \texttt{ct}_{\text{ecies}}, \sigma_{\text{cscc}}, \texttt{hash}_{\text{blk}}, \texttt{shard}_{\text{dst}}, \texttt{pool}_{\text{dst}} \rangle$.
Ensure: State transition on target shard Merkle tree and payload archival.

1: Identify execution type: $\texttt{is_shielded} == \text{true}$.
2: Consensus invokes APLE (SYSTEM_EXECUTOR, address 0x...ff).
3: Retrieve source committee public key $\texttt{pk}_{\text{src_bft}}$ from election table for $\texttt{hash}_{\text{blk}}$.
4: Compute message hash $M \leftarrow H(\text{cm}_{\text{send}} \parallel \texttt{shard}_{\text{dst}} \parallel \texttt{pool}_{\text{dst}} \parallel \texttt{hash}_{\text{blk}})$.
5: **Assert** $e(\sigma_{\text{cscc}}, G_2) == e(M, \texttt{pk}_{\text{src_bft}})$ {Single pairing check, $\mathcal{O}(1)$}
6: **Assert** $\text{cm}_{\text{send}} \notin \text{CommitmentSet}_{\text{dst}}$ {Prevent replay}
7: Insert commitment: $\text{CommitmentSet}_{\text{dst}} \leftarrow \text{CommitmentSet}_{\text{dst}} \cup \{\text{cm}_{\text{send}}\}$.
8: Update target Sparse Merkle Tree leaf: $\text{SMT}_{\text{dst}}.\text{Insert}(\text{cm}_{\text{send}})$.
9: Archive ciphertext: $\text{Store}(\text{cm}_{\text{send}}, \texttt{ct}_{\text{ecies}})$.
10: Emit event $\texttt{ShieldedCrossTransferIn}(\text{cm}_{\text{send}}, \text{SMT}_{\text{dst}}.\text{Root}())$.

G. Cross-Shard Atomicity and Fault Tolerance

In Shardora, cross-shard shielded transfers achieve **deterministic source-side finality**. Nullifier invalidation, change commitment insertion, and CSCC generation are committed within a single atomic block finalized by BFT consensus. Once finalized, the source state update is irreversible.

Target shard processing via ZPCM is idempotent:

$$\text{Assert } \text{cm}_{\text{send}} \notin \text{CommitmentTree}_{\text{dst}}. \tag{21}$$

Duplicate delivery results in a no-op. Under network partitions, CSCC messages remain queued in persistent message buffers. Upon partition recovery, target shards resume blind insertions unconditionally.

Conventional cross-shard 2-Phase Commit (2PC) mechanisms introduce privacy leaks: timeout rollbacks mandate refunding funds to the original sender address on the source shard, creating an on-chain link between the spent nullifier and the refund address. By establishing source-side finality and idempotent target commitment, Shardora eliminates timeout rollback logic, preserving zero-leakage privacy under network faults.

H. Client Scanning Optimization: 1-Byte View-Tag Filtering

Trial decryption in shielded protocols incurs client overhead. To protect resource-constrained mobile clients under high throughput, Shardora introduces a two-tier screening pipeline:

1) Denomination Pool Partitioning: DSPE shards are divided into P distinct denomination pools (e.g., $P = 4$: 0.1, 1, 10, 100 tokens). Clients subscribe only to CSSC streams corresponding to their held denomination, reducing the scanning surface by $(1 - 1/P) = 75\%$.

2) 1-Byte View-Tag: We prepend a 1-byte tag to $\texttt{ct}_{\text{ecies}}$, defined as:

$$\texttt{view_tag} = H_{\text{p}}(S)[0] \pmod{256}. \tag{22}$$

The client computes $S = \texttt{vsk} \cdot R$ via scalar multiplication and tests the tag before running symmetric decryption. Mismatches are rejected immediately (99.6% rejection rate). Consequently, less than 0.1% of total network transactions require full AES-GCM decryption, reducing client scanning cost to ≤ 16 ms/second under peak network load.

Algorithm 3 Two-Tier Trial Decryption Pipeline

Require: Viewing private key $\texttt{vsk}$, spending public key $\texttt{pk}_{\text{spend}}$, observed pool messages $\{\text{CSSC}_j\}_{j=1}^{N}$ for subscribed denomination p^*.
Ensure: Identified owned Note instances.

```
for each CSSC_j = ⟨cm_j, ct_{ecies,j}, ...⟩ do
    Extract ephemeral key R_j and tag τ_j ← ct_{ecies,j}.view_tag.
    Compute ECDH point S_j ← vsk · R_j.
    Compute test tag τ* ← H_p(S_j)[0] (mod 256).
    if τ_j ≠ τ* then
        continue {Rejected with 99.6% probability; skip AES}
    end if
    Derive k_enc ← HKDF-SHA256(S_j ‖ "shardora_note_v1").
    pt_j ← AES-256-GCM-Dec_{k_enc}(ct_{ecies,j}.payload, AAD = cm_j).
    if Decryption succeeds and pt_j.pk_stealth == H_p(S_j) · G + pk_spend then
        Record Note: OwnedNotes.Add(pt_j.v, pt_j.r, S_j).
    end if
end for
```

IV. Formal Security Theory and Proofs

A. Security Definitions and Hardness Assumptions

Assumption 1 (Discrete Logarithm (DL))**.** For group $\mathbb{G}_1$ of prime order p and generator G, $\Pr[\mathcal{A}(G, a{\cdot}G) = a] \leq \text{negl}(\lambda)$.

Assumption 2 (Computational Diffie-Hellman (CDH))**.** For generator $G \in \mathbb{G}_1$, $\Pr[\mathcal{A}(G, a \cdot G, b \cdot G) = ab \cdot G] \leq \text{negl}(\lambda)$.

Assumption 3 (Decisional Diffie-Hellman (DDH))**.** For generator $G \in \mathbb{G}_1$, tuples (G, aG, bG, abG) and (G, aG, bG, cG) are computationally indistinguishable.

Assumption 4 (q-Strong Diffie-Hellman (q-SDH))**.** Given $(G, \tau G, \tau^2 G, \ldots, \tau^q G) \in \mathbb{G}_1^{q+1}$, $\Pr[\mathcal{A} \to (c, \frac{1}{\tau + c} G)] \leq \text{negl}(\lambda)$.

Definition 3 (Cross-Shard Source Linking Game $\text{Game}^{\text{CL}}_{\mathcal{A},\Pi}(\lambda)$)**.** The game proceeds between challenger $\mathcal{C}$ and PPT adversary $\mathcal{A}$:

1) $\mathcal{C}$ initializes protocol Π with pool size K, generating notes $\{n_1, \ldots, n_K\}$.
2) $\mathcal{A}$ adaptively queries $\texttt{Spend}(n_i)$ and observes target shard state.
3) $\mathcal{A}$ selects two unspent challenge notes n_0, n_1.
4) $\mathcal{C}$ chooses bit $b \xleftarrow{\$} \{0, 1\}$, executes $\texttt{Spend}(n_b)$, generating $\langle \texttt{nul}^*, \text{cm}^*, \texttt{ct}^*, \sigma^* \rangle$.
5) $\mathcal{A}$ receives challenge tuple and full target ledger View_{dst}, outputting guess b'.

The advantage is defined as $\text{Adv}^{\text{CL}}_{\mathcal{A},\Pi}(\lambda) = |\Pr[b' = b] - 1/2|$.

B. Structural Impossibility: Gas-First Account State Machines

Definition 4 (Gas-First Account State Machine (GASM))**.** A state machine $(\mathcal{S}, \mathcal{TX}, \to)$ is a GASM if:

- State $\mathcal{S}$ includes an account balance map $\texttt{bal} : \text{Addr} \to \mathbb{N}$.
- A transaction $\texttt{tx}$ is valid only if $\texttt{bal}[\texttt{tx.sender}] \geq \texttt{tx.gas_cost}$ prior to execution.
- State transition $\mathcal{S} \xrightarrow{\texttt{tx}} \mathcal{S}'$ deducts $\texttt{tx.gas_cost}$ from $\texttt{tx.sender}$ before evaluating application logic.

Theorem 1 (Theorem A: Impossibility of Gas-Autonomous Privacy on GASM)**.** *Let Π be any protocol running on a GASM. Π cannot simultaneously satisfy:*

1) ***Sender Privacy:*** $\Pr[\mathcal{A}(\textit{Transcript}_{\textit{chain}}) \to \textit{sender}(\texttt{tx})] \leq 1/K + \textit{negl}(\lambda)$.
2) ***Recipient Gas Autonomy:*** *A recipient address B receives funds when $\texttt{bal}[B] = 0$ without pre-existing allowances.*
3) ***Decentralization:*** *No centralized entity or colluding quorum C ($|C| \leq t_{\textit{collusion}}$) observes the unshielded sender-recipient tuple.*

Proof. Suppose Π simultaneously satisfies all three properties. By Recipient Gas Autonomy, $\texttt{bal}[B][t] = 0$ for all $t < T_{\text{recv}}$. By the GASM prerequisite, B cannot initiate any transaction $\texttt{tx}$ at $t = T_{\text{recv}}$ because $\texttt{bal}[B][t] < \texttt{tx.gas_cost}$.

Thus, the transaction crediting balance to B, denoted $\texttt{TX}_{\text{credit}}$, must be initiated by some address $S \neq B$. Because GASM mandates explicit gas deduction, $S = \texttt{TX}_{\text{credit}}.\texttt{sender}$ must be recorded publicly in $\text{Transcript}_{\text{chain}}$.

Two cases arise for S:

- **Case 1 (S is the original sender A):** Then $\text{Transcript}_{\text{chain}}$ explicitly contains A, which directly violates Sender Privacy ($\Pr[\mathcal{A} \to A] = 1$).
- **Case 2 (S is an intermediary relayer R):** To ensure R does not collude, R must belong to a decentralized set. However, R must obtain authorization to relay and be reimbursed for gas. If reimbursement occurs on-chain, the funding path links back to A. If reimbursement occurs off-chain, R possesses knowledge of the (A, B) correlation, violating Decentralization.

Hence, no such protocol Π can exist on a GASM. □

Definition 5 (Protocol-Level Account (PLA))**.** An address $P \in \text{Addr}$ is a PLA if state transitions $\mathcal{S} \xrightarrow{\texttt{tx}} \mathcal{S}'$ with $\texttt{tx.sender} = P$ bypass the condition $\texttt{bal}[P] \geq \texttt{tx.gas_cost}$, executing via direct consensus scheduling.

Corollary 2 (Evasion of Theorem A via APLE)**.** *Shardora instantiates APLE as a PLA (*SYSTEM_EXECUTOR*). Because cross-shard blind commitment insertion is executed by APLE directly through consensus dispatch, Shardora is not a GASM, circumventing Theorem 1.*

C. Information-Theoretic Source Unlinkability

Assumption 5 (Ideal Synchronization Channel Model)**.** We define the ideal cross-shard model under three assumptions:

- **B1 (Uniform Channel):** Cross-shard messages are transmitted with fixed length and uniform delay.
- **B2 (No Global Passive Adversary):** The adversary does not observe physical network topology or inter-shard transit traffic.
- **B3 (Active Anonymity Pool):** Target denomination pool contains $K \geq 2$ unspent notes.

Theorem 3 (Theorem B: Target Shard Information-Theoretic Source Unlinkability)**.** *Under Assumption 5, for any computationally unbounded adversary $\mathcal{A}$ controlling all nodes in the target shard $\mathcal{V}_{dst}$, the probability of linking a newly minted commitment cm_{send} to its corresponding source nullifier $\texttt{nul}_i$ is:*

$$\Pr[\mathcal{A}(\mathit{View}_{dst}) \to \texttt{nul}_i] = \frac{1}{K}. \tag{23}$$

That is, $\mathit{Adv}^{CL}_{\mathcal{A},\Pi}(\lambda) = 0$.

Proof. By protocol construction, the Groth16 proof π and source nullifier $\texttt{nul}_i$ are processed and archived solely on the source shard $\mathcal{V}_{\text{src}}$. The message carrier arriving at the target shard consists strictly of:

$$\text{View}_{\text{dst}} = \{\text{cm}_{\text{send}}, \texttt{ct}_{\text{ecies}}, \sigma_{\text{cscc}}\}. \tag{24}$$

We evaluate the mutual information between $\texttt{nul}_i$ and each constituent:

1) **Commitment Independence:** $\text{cm}_{\text{send}} = v \cdot H + r_{\text{send}} \cdot G$. Because $r_{\text{send}} \xleftarrow{\$} \mathbb{F}_p$ is chosen uniformly at random, Pedersen commitments are perfectly hiding. For any value v, cm_{send} is uniformly distributed over $\mathbb{G}_1$, perfectly independent of $\texttt{nul}_i$. Thus, $I(\texttt{nul}_i; \text{cm}_{\text{send}}) = 0$.
2) **Ciphertext Independence:** $\texttt{ct}_{\text{ecies}}$ encrypts plaintext $(v_{\text{send}}, r_{\text{send}}, \texttt{pk}_{\text{stealth}})$. The source nullifier $\texttt{nul}_i = H_{\text{p}}(\texttt{sk}_{\text{old}} \parallel \text{cm}_{\text{old}})$ contains only old Note parameters, which do not appear in $\texttt{ct}_{\text{ecies}}$. By the data processing inequality, $I(\texttt{nul}_i; \texttt{ct}_{\text{ecies}}) \leq I(\texttt{nul}_i; \texttt{pt}) = 0$.
3) **Certificate Independence:** σ_{cscc} is a BLS signature over $(\text{cm}_{\text{send}} \parallel \texttt{shard}_{\text{dst}} \parallel \texttt{pool}_{\text{dst}} \parallel \texttt{hash}_{\text{blk}})$. None of these signed inputs contain $\texttt{nul}_i$. Since $\mathcal{V}_{\text{src}} \cap \mathcal{V}_{\text{dst}} = \emptyset$, $\mathcal{A}$ has no access to the pre-image block structure of $\texttt{hash}_{\text{blk}}$. Thus, $I(\texttt{nul}_i; \sigma_{\text{cscc}}) = 0$.

By the subadditivity of mutual information:

$$I(\texttt{nul}_i; \text{View}_{\text{dst}}) \leq I(\texttt{nul}_i; \text{cm}) + I(\texttt{nul}_i; \texttt{ct}) + I(\texttt{nul}_i; \sigma) = 0. \tag{25}$$

Because the mutual information is zero, the posterior distribution of the source index matches the uniform prior over the pool:

$$\Pr[\text{source} = i \mid \text{View}_{\text{dst}}] = \Pr[\text{source} = i] = \frac{1}{K}. \tag{26}$$

The claim holds unconditionally for computationally unbounded adversaries. □

D. Multiplicative Anonymity Set Composition

Theorem 4 (Theorem C: Multiplicative Composition over Multi-Hop Routing)**.** *Assume Assumption 5 holds across M disjoint shards, with respective active pool sizes $K_1, K_2, \ldots, K_M$. If a transaction routes sequentially across M hops with fresh blinding keys sampled at each hop, the adversary's end-to-end linking probability is bounded by:*

$$\Pr[\mathcal{A} \textit{ links In to Out}] \leq \prod_{m=1}^{M} \frac{1}{K_m} + M \cdot \mathit{negl}(\lambda). \tag{27}$$

Proof. Let H_m denote the event of correctly linking hop $m-1$ to hop m. Shard committees are chosen independently via Verifiable Random Sortition (FTS), satisfying $\mathcal{V}_i \cap \mathcal{V}_j = \emptyset$ for $i \neq j$. At each intermediate hop m, the Note is unshielded into a blind intermediate commitment with freshly sampled blinding factor $r_m \xleftarrow{\$} \mathbb{F}_p$ and distinct viewing key $\texttt{pk}_{\text{view},m}$.

By Theorem 3, conditioned on the honest majority of intermediate committees (guaranteed by BFT safety), each transition represents an independent trial where $I(\text{In}_m; \text{Out}_m) = 0$. Under the CDH assumption on ECIES ciphertexts, the sequence of games yields:

$$\begin{aligned} \Pr\left[\bigcap_{m=1}^{M} H_m\right] &= \prod_{m=1}^{M} \Pr[H_m \mid H_1, \ldots, H_{m-1}] \\ &= \prod_{m=1}^{M} \frac{1}{K_m} + \sum_{m=1}^{M} \text{Adv}^{\text{CDH}}(\lambda). \end{aligned} \tag{28}$$

Because $\sum_{m=1}^{M} \text{Adv}^{\text{CDH}}(\lambda) \leq M \cdot \text{negl}(\lambda)$, the multiplicative bound holds. □

E. Dual-Track Decoupled Security

Theorem 5 (Theorem D: Dual-Track Security Decoupling)**.** *Let λ be the security parameter, n the committee size, and $t = \lceil 2n/3 \rceil$. The security of Shardora decomposes into two mutually independent reductions:*

$$\textbf{\textit{Track 1 (Confidentiality):}} \quad Adv_{\mathcal{A}}^{conf}(\lambda) \leq Adv_{\mathbb{G}_1}^{CDH}(\lambda) + negl(\lambda), \tag{29}$$

$$\textbf{\textit{Track 2 (Integrity):}} \quad Adv_{\mathcal{A}}^{integ}(\lambda) \leq Adv_{\mathcal{A}}^{BFT\text{-}safety}(n,t) + Adv_{BLS}^{EUF\text{-}CMA}(\lambda). \tag{30}$$

Proof. **Track 1:** Note confidentiality requires determining plaintext $\mathtt{pt} = (v, r, \mathtt{pk}_{\text{stealth}})$ from $\mathtt{ct}_{\text{ecies}} = (R, \mathtt{ct})$. The decryption key is derived via $S = \mathtt{vsk} \cdot R$. The viewing private key $\mathtt{vsk}$ is held exclusively by the recipient and is never shared with validator committees or transmitted across shards. By reduction to ECIES in the Random Oracle Model (ROM) [10], any algorithm distinguishing $\mathtt{ct}_{\text{ecies}}$ can be transformed into a solver for the CDH problem in $\mathbb{G}_1$. This reduction is independent of committee size n, Byzantine fraction f, or consensus liveness.

Track 2: Ledger integrity requires that no invalid commitment cm^* (i.e., created without valid consumed balance) is minted on the target shard. A target shard executes ZPCM only if provided with a valid certificate σ_{cscc}. Forging σ_{cscc} requires either:

1) Breaking the existential unforgeability under chosen message attacks (EUF-CMA) of the BLS signature scheme on $\mathbb{G}_2$.
2) Corrupting at least $t = \lceil 2n/3 \rceil$ validators in the source committee to sign an unproven state transition, violating base BFT safety.

Summing the probabilities of these disjoint events yields the Track 2 bound. □

Lemma 6 (Partial Byzantine Corruption Behavior)**.** *Let f' be the number of Byzantine nodes in a committee of size n, where threshold $t = \lceil 2n/3 \rceil$. For $n/3 \leq f' < 2n/3$:*

$$Adv_{\mathcal{A}}^{integ}(f', n) \leq Adv_{BLS}^{EUF\text{-}CMA}(\lambda) + Adv_{ZK}^{Soundness}(\lambda) + negl(\lambda). \tag{31}$$

Integrity is strictly maintained, while consensus liveness may degrade if Byzantine nodes refuse to sign.

Corollary 7 (Consensus Failure Resilience)**.** *Even in the scenario where 100% of consensus nodes become Byzantine ($Adv^{BFT\text{-}safety} = 1$), Note confidentiality (Track 1) remains bounded by Adv^{CDH}. Adversaries cannot decrypt historical or inflight Note payloads.*

F. Privacy-Liveness Compatibility

Theorem 8 (Theorem E: Privacy-Liveness Compatibility)**.** *Assuming standard BFT liveness ($< 1/3$ Byzantine nodes), every valid shielded cross-shard transaction is finalized within $\mathcal{O}(\kappa \cdot \Delta)$ time with probability $1 - negl(\lambda)$, where $\kappa = n/(n-t)$ and Δ is the network delay bound.*

Proof. By HotStuff consensus properties, a valid proof submitted to an honest source leader is included within $\mathcal{O}(\kappa \cdot \Delta)$. Generation of σ_{cscc} is embedded directly into the block proposal and vote rounds. Routing across the network layer incurs bounded delay under partial synchrony. Target execution via ZPCM requires only local BLS signature verification and SMT insertion, which executes deterministically in $\mathcal{O}(1)$ without multi-phase blocking. □

G. End-to-End Cross-Shard Privacy Reduction

Theorem 9 (Theorem 3: End-to-End Shielded Secrecy)**.** *Under the CDH, DL, q-SDH, and ROM assumptions, for any polynomial-time observer $\mathcal{O}$ and colluding adversary corrupting $< 1/3$ nodes in each shard, the Shardora protocol satisfies sender k-anonymity, recipient untraceability, amount confidentiality, and cross-shard unlinkability simultaneously.*

Proof. We construct a sequence of hybrid experiments $H_0 \ldots H_4$:

- H_0: Real execution of protocol Π.
- H_1: Replace ECIES ciphertext $\mathtt{ct}_{\text{ecies}}$ with uniform random bits. By Theorem 1a, $|\Pr[H_1] - \Pr[H_0]| \leq \text{Adv}_{\mathbb{G}_1}^{\text{CDH}}(\lambda)$.
- H_2: Replace Groth16 proof π with simulated proof π_{sim} using zk-SNARK zero-knowledge simulator. $|\Pr[H_2] - \Pr[H_1]| \leq \text{negl}(\lambda)$.
- H_3: Replace Pedersen commitments $\text{cm}_{\text{send}}, \text{cm}_{\text{change}}$ with uniformly chosen elements of $\mathbb{G}_1$. By perfect hiding of Pedersen commitments, $|\Pr[H_3] - \Pr[H_2]| = 0$.
- H_4: Replace nullifier $\mathtt{nul}$ with an independent random field element from $\mathbb{F}_p$. By ROM evaluation of H_{p}, $|\Pr[H_4] - \Pr[H_3]| \leq \text{negl}(\lambda)$.

In H_4, all view components are statistically independent of witness values, establishing the bound. □

H. Real-World Network Degradation Analysis

In real deployments, Assumption 5 can be challenged by network side channels. We model three major side channels and formalize the composite degradation bound.

1) Timing Side Channels: Let $\tau_{\min}$ be the adversary's timestamping precision, and μ the parameter of an exponential random transit delay $\Delta_{\text{delay}} \sim \text{Exp}(\mu)$.

Lemma 10 (Lemma F.1: Timing Degradation)**.** *Let N_W denote the expected number of candidate transactions in a window of width $3/\mu$. The adversary's linking advantage from timing metadata is:*

$$\varepsilon_{timing}(\mu) = \frac{N_W}{K} \cdot (1 - e^{-\mu \tau_{\min}}). \tag{32}$$

Proof. For any candidate transaction $i' \neq i^*$, the probability that its arrival time falls within the adversary's indistinguishability window $\tau_{\min}$ of the target transaction is given by $\Pr[\Delta \leq \tau_{\min}] = 1 - e^{-\mu \tau_{\min}}$. Applying the union bound over N_W competing candidate transfers yields the stated formulation. □

2) Global Passive Adversary (GPA) Traffic Analysis: Let an AS-level observer monitor source and target shard traffic volumes M_{src} and M_{dst} over epoch W. Let H_s denote the message size distribution entropy, and $H_t = W/\tau_{\min}$ the temporal bin entropy.

Lemma 11 (Lemma F.2: GPA Traffic Correlation Advantage)**.** *The maximum a posteriori (MAP) linking advantage of a GPA is:*

$$\varepsilon_{GPA} \leq \frac{M_{src} \cdot M_{dst}}{K^2 \cdot B^2 \cdot H_s \cdot H_t}, \tag{33}$$

where B is the transaction batch aggregation factor.

Proof. The GPA performs bipartite matching between source outputs and target inputs. The probability that an arbitrary incorrect source transaction matches both size and time buckets of the target is $p_s \cdot p_t$, where $p_s = M_{\text{dst}}/(KBH_s)$ and $p_t = M_{\text{src}}/(KBH_t)$. Multiplying these matching probabilities bounds the maximum advantage over uniform guessing. □

3) Amount Graph Analysis: Let n_{pool} be the denomination pool size, and q the number of publicly observed unshielding events.

Lemma 12 (Lemma F.3: Amount Graph VC Bound)**.** *For an adversary utilizing a hypothesis class $\mathcal{H}$ of single-dimensional amount threshold classifiers (VC($\mathcal{H}$) ≤ 2), the linking advantage is bounded by:*

$$\varepsilon_{amount} \leq \sqrt{\frac{q \ln(2/\delta)}{2n_{pool}}}, \tag{34}$$

with probability at least $1 - \delta$.

Proof. Direct application of the Vapnik-Chervonenkis (VC) inequality for empirical risk minimization over linear threshold concepts on $\mathbb{R}^+$. □

4) Composite Real-World Degradation Theorem:

Theorem 13 (Theorem F: Real-World Degradation Bound)**.** *Let protocol Π operate over a physical network with mitigation stack $\mathcal{M} = (pad_{1024B}, Exp\text{-}delay_{\mu}, batch_B)$. Under timing, GPA, and amount graph adversaries, the composite linking probability is bounded by:*

$$\Pr[\mathcal{A} \text{ links } \texttt{nul}_i \to cm_j] \leq \frac{1}{K} + \varepsilon_{real}, \tag{35}$$

where by subadditivity of conditional mutual information:

$$\varepsilon_{real} = \varepsilon_{timing} + \varepsilon_{GPA} + \varepsilon_{amount}. \tag{36}$$

Proof. Let X denote the source index random variable ($H(X) = \log K$), and let $Y_0 = \text{View}_{\text{dst}}$, $Y_1 =$ Timing, $Y_2 =$ Traffic, and $Y_3 =$ Amount. The total mutual information available to the adversary is:

$$\begin{aligned} I(X; Y_0, Y_1, Y_2, Y_3) &= I(X; Y_0) + I(X; Y_1, Y_2, Y_3 \mid Y_0) \\ &= 0 + I(X; Y_1, Y_2, Y_3 \mid Y_0), \end{aligned} \tag{37}$$

since $I(X; Y_0) = 0$ by Theorem 3. Applying the chain rule and subadditivity of conditional mutual information:

$$I(X; Y_1, Y_2, Y_3 \mid Y_0) \leq I(X; Y_1) + I(X; Y_2) + I(X; Y_3). \tag{38}$$

Applying Lemmas 10, 11, and 12 to each component yields the additive bound. □

Corollary 14 (Concrete Parameter Bounds)**.** *For parameters $K = 1,000$, $\mu^{-1} = 300$ s, $B = 50$, $\tau_{\min} = 100$ ms, constant padding $s = 1024$ Bytes ($H_s \to \infty \implies \varepsilon_{GPA} = 0$), $n_{pool} = 10^7$, and $q \leq 10$, the composite degradation satisfies:*

$$\begin{aligned} &\varepsilon_{timing} \approx 1.67 \times 10^{-5}, \quad \varepsilon_{GPA} = 0, \quad \varepsilon_{amount} \approx 5.8 \times 10^{-4}, \\ &\varepsilon_{real} \leq 5.96 \times 10^{-4} < 10^{-3}. \end{aligned} \tag{39}$$

The real-world linking advantage is bounded by $\leq 1/1000 + 0.0006 \approx 0.0016$.

I. Trusted Setup and PLONK Migration Analysis

Groth16 requires a circuit-specific common reference string (CRS), created through a multi-party computation (MPC) ceremony. For Shardora's circuit size ($|\mathcal{C}| = 21,010$), the Phase 2 ceremony requires ~ 100 participants and finishes within 2–4 weeks.

If trusted setup requirements present an adoption barrier, Shardora supports migration to universal SRS systems such as PLONK [11]:

- **Proof Size:** Increases from 256 B to ≈ 800 B ($+212\%$).
- **Verification Overhead:** Verification cost increases from $\approx$ 3 ms to ≈ 8 ms.
- **Throughput Impact:** Estimated single-pool throughput decreases from $1,265$ to $\approx 1,100$ TPS ($\approx -13\%$).

Because circuit constraints C1–C10 are expressed in standard R1CS arithmetic, transitioning to PLONK requires changing only the prover/verifier backend contracts, leaving cross-shard protocol mechanics unchanged.

V. User Shield/Unshield Lifecycle and Mechanics

A. Dual-Domain State Representation

User assets exist across two distinct ledger domains within each shard:

1) **Transparent Domain:** Standard account-based mapping (`_balances[address]` → uint256).
2) **Shielded Domain:** UTXO-based commitment state managed by the DSPE Sparse Merkle Tree (`cm_tree`).

B. Shield and Unshield Formal Security

Definition 6 (Shield Operation Security)**.** A shield operation $\texttt{Sh}(v, \texttt{pk}_{\text{spend}}, r) \to$ cm takes public amount v, public key $\texttt{pk}_{\text{spend}}$, and random scalar $r \xleftarrow{\$} \mathbb{F}_p$, appending $\text{cm} = v \cdot H + r \cdot G$ to the SMT.

Definition 7 (Unshield Operation Security)**.** An unshield operation $\texttt{Unsh}(\texttt{sk}, \mathcal{P}, \texttt{addr}_{\text{recv}}, v) \to (\texttt{nul}, \pi)$ publishes nullifier $\texttt{nul} = H_{\text{p}}(\texttt{sk} \parallel \text{cm})$ and transfers transparent balance v to $\texttt{addr}_{\text{recv}}$.

Theorem 15 (Theorem 4: Post-Shield Indistinguishability)**.** *In the Random Oracle Model under the DDH assumption, for an SMT containing k leaves, the advantage of distinguishing whether a Note was minted to $\texttt{pk}_0$ or $\texttt{pk}_1$ is bounded by*

negl(λ)*, and associating subsequent spent nullifiers with the deposit event is bounded by:*

$$\Pr[\mathcal{A} \textit{ associates } \texttt{nul} \to \texttt{Deposit}] \leq \frac{1}{k} + \textit{negl}(\lambda). \quad (40)$$

Proof. Pedersen commitments are perfectly hiding over uniformly random blinding factor $r \in \mathbb{F}_p$. Nullifier derivation $\texttt{nul} = H_{\text{p}}(\texttt{sk} \parallel \text{cm})$ constitutes a one-way evaluation in the ROM. For an adversary without sk, nul is computationally indistinguishable from a random field element, limiting association to random guessing among k leaves. □

Theorem 16 (Theorem 5: Unshield Exposure Bound)**.** *Any valid unshield operation reveals* $\texttt{addr}_{recv}$ *and amount* v *on-chain. However, under the DL, q-SDH, and ROM assumptions, the spending key* sk *and the specific source commitment cm remain computationally hidden:*

$$\Pr[\mathcal{A}(\texttt{Unsh}) \to \textit{cm}_{spent}] \leq \frac{1}{K} + \textit{negl}(\lambda). \quad (41)$$

Proof. The transparent balance transfer requires specifying $\texttt{addr}_{\text{recv}}$ and v. The Groth16 proof π proves that some leaf in the SMT satisfies value conservation without revealing the leaf index. By zero-knowledge of Groth16, π can be simulated without witness access. The spent leaf index remains concealed among the K candidate leaves. □

Corollary 17 (Optimal Unshielding Strategy)**.** *To prevent correlation via amount matching, users should: (1) shield using standard denominations, (2) perform* ≥ 2 *cross-shard hops, and (3) apply an exponential randomized delay* $\Delta_t \sim \textit{Exp}(1/300)$ *prior to unshielding.*

Theorem 18 (Theorem 6: End-to-End Hybrid Lifecycle Privacy)**.** *Under the DDH, DL, q-SDH, and ROM assumptions, across a complete lifecycle of* $\texttt{Shield} \to \texttt{CrossShardTx}^M \to \texttt{Unshield}$*, the information revealed to a polynomial-time observer* $\mathcal{A}$ *is bounded by:*

$$I_{\mathcal{A}} = \{v_{shield}, t_{shield}, v_{unshield}, \texttt{addr}_{recv}, t_{unshield}\}. \quad (42)$$

The advantage in linking the source deposit to the unshielded withdrawal is bounded by:

$$Adv_{\mathcal{A}}^{lifecycle} \leq \prod_{m=1}^{M} \frac{1}{K_m} + \varepsilon_{real} + \textit{negl}(\lambda). \quad (43)$$

Proof. Combines Theorems 3, 4, 13, and Theorem 5 using standard hybrid sequence arguments. □

C. Client SDK Architecture and Security Tiers

The Shardora Wallet SDK provides three configurable privacy tiers:

- **PRIVACY_HIGH:** Batch size $B = 50$, delay $\mu^{-1} = 300$ s, padding 1024 B, multi-hop $M = 2$. Effective linking advantage $\leq 1/K + 10^{-3}$.
- **PRIVACY_BALANCED:** Batch size $B = 10$, delay $\mu^{-1} = 30$ s, padding 1024 B, single hop $M = 1$. Effective linking advantage $\leq 1/K + 10^{-2}$.
- **PRIVACY_FAST:** Immediate dispatch ($B = 1$, zero delay), padding 1024 B. Recommended only for low-value transfers.

VI. Quantitative Performance Analysis

A. System Parameter Baseline

Performance evaluations are established using constants configured in branch xl0616 (Table VI).

B. End-to-End Latency Breakdown

Table VII presents the step-by-step latency comparison between transparent and shielded cross-shard transfers. Shielded transfers add less than 8% latency overhead over transparent operations.

C. Pairing Bottleneck Elimination

If target shards were required to verify full Groth16 proofs, processing 1,200 transactions per block would require $3 \times 1,200 = 3,600$ BN254 pairings (5.4 s computation), exhausting block execution capacity.

Under Shardora's CSCC model, target shards execute **zero Groth16 pairings**:

$$T_{\text{target_block}} = T_{\text{BLS_verify}} + B_{\text{tx}} \cdot T_{\text{SMT_insert}}, \quad (44)$$

where $T_{\text{BLS_verify}} \approx 1.8$ ms (a single aggregate pairing check for the entire block) and $T_{\text{SMT_insert}} \approx 0.04$ ms. For $B_{\text{tx}} = 1,000$, target validation completes in < 45 ms.

D. Throughput Projections

On the source shard, pairing verification overhead is mitigated via Bellman batch verification:

$$e\left(\sum_{j=1}^{m} \alpha_j A_j, B_j\right) = \cdots \implies 2 + m \text{ pairings}, \quad (45)$$

reducing pairing verification time by 64% for batches of $m = 50$. Table IX presents projected throughput scaling across different verification modes.

E. Proving Engine Benchmarks

Proving times for the 21,010-constraint circuit across execution environments are:

- **Desktop CPU (Rust/arkworks, 8 cores):** 1.2–1.8 s.
- **GPU Prover (CUDA, RTX 4090):** ≈ 180 ms.
- **Mobile Client (ARM Cortex-A78):** 8.5–14.0 s.

Mobile clients can delegate proof generation to untrusted proving services: delegating witness generation without the spending key sk protects secret keys, since the nullifier $\texttt{nul} = H_{\text{p}}(\texttt{sk} \parallel \text{cm})$ can be computed locally prior to delegation.

F. Empirical Validation Gap Statement

We clarify that the performance numbers reported in Table IX and Table VIII are analytical projections derived from benchmarked BN254 operation costs on server-grade hardware. Full empirical stress testing under wide-area distributed network conditions with adversarial churn remains future work.

VII. Implementation, Roadmaps, and Discussion

A. Implementation Mapping

The Shardora shielded transaction pipeline is prototyped within the active system branch (`xl0616`). Table X maps protocol components to specific codebase modules.

B. Compatibility and Deployment

1) Protobuf Backward Compatibility: All shielded extensions in `ToTxMessageItem` (`new_commitment`, `ecies_ct`, `cscc_signature`) are defined as optional protobuf fields. Legacy nodes ignore unknown tags and process standard transactions without modification.

2) Phased Activation: To avoid network splits, validator state records incorporate a boolean flag `privacy_enabled`. Proposals containing `is_shielded` $=$ true transactions are rejected until 100% of committee members confirm support, matching standard EIP activation patterns.

C. Phased Implementation Roadmap

- **Phase 1 (Recipient Privacy):** Implement stealth address generation; add ephemeral public key tracking to the contract.
- **Phase 2 (Value Confidentiality):** Introduce Pedersen commitments and Poseidon SMTs; deploy intra-shard shielded transfers with Groth16.
- **Phase 3 (Cross-Shard Privacy):** Deploy CSCC BLS aggregation, APLE execution hooks, and ZPCM blind commitment minting.
- **Phase 4 (Anonymity Enhancements):** Establish standardized denomination sub-pools and default delay routing in client SDKs.

D. Post-Quantum Considerations

BN254 elliptic curve operations are vulnerable to polynomial-time discrete logarithm evaluation via Shor's algorithm. Long-term post-quantum upgrades require replacing BN254 primitives: Groth16 with lattice-based SNARKs (e.g., Latticefold), ECIES with ML-KEM (CRYSTALS-Kyber), BLS with ML-DSA (CRYSTALS-Dilithium), and Pedersen commitments with hash-based commitments. While proof sizes currently present engineering challenges, the structural separation between consensus authorization and zero-knowledge proof domains isolates cryptographic upgrades to individual components.

VIII. Conclusion

While single-chain state machines can host privacy-preserving mixers, they remain structurally vulnerable to gas correlation, global observation collapse, and relayer trust expansion. Cross-shard architectures resolve these challenges fundamentally.

In this work, we presented the complete architecture of Shardora. We proved the impossibility of gas-autonomous privacy on standard account state machines, demonstrated that Shardora evades this result via APLE, and proved that physical observation domain separation provides information-theoretic source unlinkability under disjoint validator quorums. We introduced dual-track security decoupling, modeled side-channel degradation in physical networks, and demonstrated that cross-shard credit certificates eliminate pairing operations on target shards, establishing that sharded consensus networks can provide strong structural privacy guarantees alongside horizontal scalability.

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TABLE I
SYSTEMATIC COMPARISON OF PRIVACY PROTOCOLS ACROSS N1, N2, AND OPERATIONAL DIMENSIONS

Protocol	**Sender Anon.**	**Recipient Priv.**	**Confidentiality**	**Property N1**	**Property N2**	**Gas Auton.**	**Proof System**	**Core Structural Limitation**
Tornado Cash	✓	✓	Fixed Denom.	×	N/A	×	Groth16	Same-chain visibility; Gas linkage unavoidable.
Zcash Sapling	✓	✓	✓	×	N/A	✓	Groth16	Monolithic L1; no cross-shard domain separation.
Monero	✓	✓	✓	×	N/A	✓	Bulletproofs	Ring signature decoys scale poorly; no EVM support.
Aztec v2	✓	✓	✓	△	×	△	PLONK	Public L1 bridge events; centralized sequencer.
Penumbra (IBC)	✓	✓	✓	△	×	△	Groth16	IBC relayer observes plaintext routing metadata.
Railgun / Bridge	✓	✓	✓	×	×	×	Groth16	Public lock/mint events form linkable on-chain

TABLE II
MATHEMATICAL COMPARISON OF PROOF SYSTEMS

System	Proof Size	Prover	Verifier	Setup	Recursion
Groth16	~ 256 B	$\mathcal{O}(\lvert\mathcal{C}\rvert \log \lvert\mathcal{C}\rvert)$	$\mathcal{O}(1)$ (3 pair.)	Circuit	$\times$
PLONK	~ 800 B	$\mathcal{O}(\lvert\mathcal{C}\rvert \log \lvert\mathcal{C}\rvert)$	$\mathcal{O}(1)$ (pairing)	Universal	✓
Halo2	~ 1.5 KB	$\mathcal{O}(\lvert\mathcal{C}\rvert \log \lvert\mathcal{C}\rvert)$	$\mathcal{O}(\log \lvert\mathcal{C}\rvert)$	Transparent	✓
Bulletproofs	$\mathcal{O}(\log \lvert\mathcal{C}\rvert)$	$\mathcal{O}(\lvert\mathcal{C}\rvert)$	$\mathcal{O}(\lvert\mathcal{C}\rvert)$	Transparent	$\times$
STARKs	$\mathcal{O}(\log^2 \lvert\mathcal{C}\rvert)$	$\mathcal{O}(\lvert\mathcal{C}\rvert \log \lvert\mathcal{C}\rvert)$	$\mathcal{O}(\log^2 \lvert\mathcal{C}\rvert)$	Transparent	✓

TABLE III
R1CS CIRCUIT COMPLEXITY BREAKDOWN

Sub-Circuit Component	Primitive	R1CS Constraints
Pedersen Commitments ($\times 3$)	Variable Base Scalar	750
Merkle Path (Depth 20)	Poseidon Hash	10,000
Public Key Derivation	Fixed Base Scalar Mul	4,000
Nullifier Derivation	Poseidon ($2 \to 1$)	500
Value Conservation	Linear Addition	10
Range Proofs (2×64-bit)	Bit Decomposition	4,000
Output Commitments & Binding	Linear EC Comb.	1,750
Total Constraint Count	$\lvert\mathcal{C}\rvert$	**21,010**

TABLE IV
REAL-WORLD NETWORK SIDE-CHANNEL DEGRADATION AND MITIGATION MATRIX

Side Channel	**Attack Vector**	**Degraded** $\Pr[\textbf{link}]$	**Engineering Mitigation**	**Residual Advantage**
Packet Size Variations	CSSC payload differences leak Note denominations	$1/K'$ ($K' \ll K$)	Pad all messages to fixed 1024 Bytes	$1/K$ (Restored)
Traffic Graph Analysis	GPA measures cross-shard volumes M_s, M_d	$M_s M_d/(K^2 H_s H_t)$	Batching $B = 50$ + 10% dummy CSSC traffic	$\leq 10^{-7}$ (Negligible)
Timing Correlation	Low network volume isolates inter-shard arrivals	$1/N_\delta$	Random transit delays $\Delta \sim \text{Exp}(1/300\text{ s})$	$\leq 1.67 \times 10^{-5}$
Shared Cloud Provider	Co-located validator instances in same AWS region	Equiv. to GPA	Enforce validator diversity ($< 1/3$ per cloud)	Falls back to GPA model
Amount Graph Matching	Matching deposit/withdrawal pairs across pools	$1/K + \sqrt{q \ln 2/(2n_{\text{pool}})}$	Fixed denomination sub-pools ($P = 4$)	$\leq 5.8 \times 10^{-4}$

TABLE V
ANONYMITY SET DEGRADATION SIMULATION ($K = 1,000$, WINDOW 1H)

Scenario	Behavioral Profile	Effective K_{eff}	Degradation Source
Optimal	Standard denom., $M = 2$, delayed	$\approx 1,000$	Minimal
Typical	Uniform amount, zero delay	≈ 100	Timing side-channel
Pathological	Unique amount, immediate exit	1	Amount correlation
Low Load	1 transaction per minute	≈ 5	Small temporal window
High Load	100 transactions per minute	≈ 980	Temporal confusion

TABLE VI
SYSTEM AND CONSENSUS PARAMETER BASELINE

Parameter Name	Symbol	Configured Value
Leader Rotation Epoch	T_{leader}	10 s
Epoch Duration	T_{epoch}	600 s
Max Block Transaction Limit	$N_{\text{tx_max}}$	20,480
Max Block Size	$S_{\text{block_max}}$	1 MB
Parallel Pools per Shard	P_{shard}	32
Committee Size	n	100 ($t = 67$)

TABLE VII
CROSS-SHARD LATENCY BREAKDOWN

Execution Phase	Shielded Transfer	Transparent Transfer
Client Proof Gen. (CPU)	1.2 s (Off-critical)	0 s
Source Shard Consensus	10.0 s	10.0 s
CSCC BLS Aggregation	0.002 s	0 s
Inter-Shard GBP Routing	5.0 s	5.0 s
Target ZPCM Verification	0.002 s	0.001 s
Target Shard Consensus	10.0 s	10.0 s
Total E2E Latency	**26.2** s	**25.0** s

TABLE VIII
TARGET SHARD EXECUTION COST (1,000 TRANSACTIONS)

Operation Phase	Naive Dual-ZK	Shardora (CSCC)
Pairing Computations	5,400 ms (3,000 pairings)	**1.8** ms (1 aggregate)
SMT State Insertion	40.0 ms	40.0 ms
BLS Verification	0.0 ms	1.8 ms
Total Verification Time	**5,440.0** ms	**43.6** ms

TABLE IX
ANALYTICAL THROUGHPUT (TPS) ACROSS CONFIGURATIONS

Verification Strategy	Pool TPS	Shard TPS	10-Shard TPS
Standard Verification (Base)	15	480	4,800
Batch Pairing Verification	50	1,600	16,000
GPU SnarkPack Aggregation	100+	3,200+	32,000+

TABLE X
SYSTEM MODULES AND INTEGRATION MATRIX

Component	Source File	Functional Role
DSPE Engine	`privacy_shadow.sol`	SMT state, Groth16 EVM verifier.
EVM Logging	`shardora_host.cc`	Intercepts `CrossTransferOut`.
CSCC Attest	`contract_call.cc`	Appends BLS signature σ_{cscc}.
Proto Schema	`pools.proto`	Defines `ToTxMessageItem` fields (21–26).
ZPCM Engine	`to_tx_local_item.cc`	Executes `ShieldedCreditFromCSCC`.
Crypto Core	`bls_dkg.cc`	BN254 pairings and BLS verification.

TABLE XI
TAXONOMY OF CORE THEORETICAL RESULTS AND CRYPTOGRAPHIC PROOFS

Result	**Scope**	**Security Type**	**Underlying Assumption**	**Core Significance**
Theorem A	Monolithic state machines	Impossibility	Standard GASM model	Proves Gas-autonomous privacy impossible without PLA/APLE.
Theorem B	Sharded blockchains	Information-Theoretic	Uniform channel, disjoint quorums	Establishes exact $1/K$ linking upper bound on destination shard.
Theorem C	Multi-hop routing	Computational	CDH, disjoint quorums	Proves multiplicative anonymity compounding $\prod 1/K_i$.
Theorem D	Cross-shard carrier	Decoupled / Computational	CDH $\vee$ (BFT safety $\wedge$ EUF-CMA)	Proves privacy persists even upon total consensus collapse.
Theorem E	Liveness / Consensus	System Liveness	BFT Liveness ($f < n/3$)	Guarantees deterministic termination within $\mathcal{O}(\kappa\Delta)$ rounds.
Theorem F	Real-world networks	Quantified Degradation	Empirical channel model	Explicitly bounds real-world side-channel degradation to $< 10^{-3}$.
Theorem 1a	Encryption payload	Computational IND-CCA2	CDH in $\mathbb{G}_1$, ROM	Ensures Note secrecy against intermediate routing nodes.
Theorem 1b	Credit certificate	EUF-CMA Unforgeability	co-CDH in $\mathbb{G}_2$	Prevents unauthorized injection of counterfeit commitments.
Theorem 4	Deposit boundary	Computational Indistinguishability	DDH, ROM	Hides recipient identity during transition to shielded pool.
Theorem 5	Withdrawal boundary	Computational Zero-Knowledge	q-SDH, ROM	Conceals ancestral commitment path during unshielding.
Theorem 6	Full transaction lifecycle	Composite Reduction	DDH, DL, q-SDH, ROM	Guarantees end-to-end zero-knowledge lifecycle privacy.